

Narges Rezaie

Data Scientist, Bioinformatics Engineer, Computational Biologist, Data Analyst, AI/ML Engineer

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Summary

Bioinformatics Engineer & Ph.D. Computational Biologist (UC Irvine) with 9+ years of experience combining scientific rigor with software engineering to build scalable bioinformatics solutions. Expertise in developing and productionizing pipelines for long- and short-read sequencing (ONT, PacBio, Illumina), single-cell multi-omics, whole-genome sequencing, metagenomics, and variant calling. Skilled in Nextflow, Snakemake, and cloud-native AWS deployments, with strong experience in benchmarking, method evaluation, workflow automation, and translating complex biological data into reliable, customer-facing tools.

Work Experience

Bioinformatics Pipeline Engineer

CA, USA

Plasmidsaurus

Dec 2024 – Present

- Own the end-to-end development and productionization of bioinformatics pipelines, serving as the primary lead for whole genome sequencing services and RNA-seq product support.
- Architect scalable workflow systems using Nextflow and Snakemake across AWS Batch and HPC environments, ensuring reproducible results for high-throughput sequencing data.
- Lead the launch of small variant calling services and Illumina metagenomics; benchmarked and integrated advanced tools for taxonomic profiling and host-read removal to optimize turnaround time and result accuracy.
- Optimize pipeline performance by resolving complex biological, sequencing, and infrastructure edge cases from initial data ingestion to the final customer deliverable.
- Drive platform evolution as a full-stack contributor, developing backend logic and frontend features that enhance data visualization and the seamless delivery of genomic results.
- Pioneer the integration of LLM and Agent-based systems to automate bioinformatics workflows, streamline complex data analysis, and elevate internal developer tooling.
- Collaborate cross-functionally with scientists, engineers, and product teams to translate novel computational methods into robust, customer-facing bioinformatics products.

Research Experience

Postdoctoral fellow

CA, USA

Mortazavi Lab, University of California, Irvine

March 2024 – Jan 2025

- Developed, applied, and optimized bioinformatic algorithms, tools, and workflows to analyze NGS datasets including short- and long-read RNA-seq at bulk and single-cell resolution, as well as Multiome, Perturb-seq, and DNA methylation data.
- Characterized the impact of genomic variants, regulatory elements, and genes on molecular and cellular phenotypes by analyzing naturally occurring or designed genomic perturbations across various cellular models ([Publication](#)).
- Identified active regulatory elements and genes for individual cell types and states by analyzing single-cell datasets.
- Gene program evaluation ([GitHub](#)): Implemented a flexible framework to evaluate the plausibility of gene programs inferred from single-cell genomic data (scRNA-seq, scATAC-seq, Multiome, and Perturb-seq) using computational methods.

Graduate Researcher

CA, USA

Mortazavi Lab, University of California, Irvine

Sept 2019 – March 2024

- Model-AD ([GitHub](#)): Analyzed diverse genomics datasets (bulk, single-cell, and single-nucleus) to characterize next-generation animal models for Alzheimer's disease and develop predictive models for therapeutic discovery. Identified gene and transcript expression changes in Model-AD mice relative to wild-type mice and human AD patients. Implemented standardized data acquisition to Synapse.
- IGVF: Investigated how genomic variation affects genome function and phenotype across tissues using single-cell data.

- **ENCODE4**: Integrated and interpreted **single-cell omics data** to profile **full-length transcriptomes** in humans and mice.
- **Gene Regulatory Networks**: Built models of **gene regulatory networks** using **bulk and single-cell RNA-seq and ATAC-seq** data in humans, rodents, and other vertebrates to study how **developmental logic** is encoded in the genome.
- **PyWGCNA** ([GitHub](#)): Developed a **Python package** for **weighted correlation network analysis (WGCNA)**, enabling **comparison of networks across datasets** and against external gene lists using **Fisher's exact test** to assess module functional enrichment.
- **Topyfic** ([GitHub](#)): Developed a **Python package** to identify **gene programs (topics)** using a robust **unsupervised machine learning** approach based on **Latent Dirichlet Allocation (LDA)**, and annotated gene programs using internal and external resources.
- **Benchmarked doublet detection** methods on **single-nucleus RNA sequencing (snRNA-seq)** datasets.
- Collaborated with **experimental scientists** to support research goals and mentored **undergraduate students** in programming and **bioinformatics pipelines**.

Education

PhD in Mathematical, Computational, and Systems Biology
[University of California, Irvine](#)

USA
 2019–2024

- **Thesis**: Computational approach to characterize gene dynamics using bulk and single-nucleus RNA sequencing to study Alzheimer's disease. **Supervisor**: Dr. Ali Mortazavi. **GPA**: 3.98/4.0

B.S. in Information Technology in Computer Engineering
[Sharif University of Technology](#)

Iran
 2014–2018

Skills

- **Bioinformatics & Computational Biology**: Single-cell/Single-nucleus RNA-seq, Multi-omics integration (scATAC-seq), Spatial Omics, Genome assembly (Hifiasm, Flye), Variant calling (Clair3), Metagenomics, and Benchmarking.
- **Pipeline Engineering & Cloud**: Production-grade **Nextflow (DSL2)** and **Snakemake** development; **AWS** ecosystem (Batch, S3, EC2, Lambda); Containerization (**Docker**); **CI/CD** (GitHub Actions); High-Performance Computing (**SLURM**).
- **Software Development**: **Python (Expert)**, R, Bash, SQL, C/C++; Web frameworks (**FastAPI**, **Pydantic**); Software testing (**Pytest**); Collaborative development (**Git**).
- **AI & Machine Learning**: **LLM-based Agentic Workflows**, Vector Databases, PyTorch, Scikit-learn, Unsupervised Learning (LDA, WGCNA), and Statistical Modeling.
- **Data Science Stack**: **Polars** (high-performance dataframes), Pandas, NumPy, SciPy, **AnnData**, Biopython, Matplotlib.
- **Domain Tools**: Seurat, Scanpy, Cellbender, Kallisto, Samtools, Bedtools, DEseq2/EdgeR.
- **Developed Packages**: **PyWGCNA**, **Topyfic**.

Selected Publications

- Identification of robust cellular programs using reproducible LDA that impact sex-specific disease progression in different genotypes of a mouse model of AD. **N Rezaie**, E Rebboah, et al. *Briefings in Bioinformatics* (2025).
- The ENCODE mouse postnatal developmental time course identifies regulatory programs of cell types and cell states. E Rebboah, **N Rezaie**, et al. *Cell Genomics* (2025).
- Long-read sequencing transcriptome quantification with Ir-kallisto. RK Loving, ..., **N Rezaie**, ..., L Pachter. *PLoS Computational Biology* 21(12): e1013692 (2025).
- BIN1^{K358R} suppresses glial response to plaques in mouse model of Alzheimer's disease. LF Garcia-Agudo, ..., **N Rezaie**, ..., KN Green. *Alzheimer's & Dementia* (2025).
- The Abca7^{V1613M} variant reduces A β generation, plaque load, and neuronal damage. CA Butler, ..., **N Rezaie**, ..., GR MacGregor, KN Green. *Alzheimer's & Dementia* (2025).
- The ENCODE4 long-read RNA-seq collection reveals distinct classes of transcript structure diversity. F Reese, ..., **N Rezaie**, et al. *Nature Methods* (2024).
- Deciphering the impact of genomic variation on function. IGVF Consortium. *Nature* (2024).

- A Trem2^{R47H} mouse model without cryptic splicing drives age-and disease-dependent tissue damage and synaptic loss in response to plaques. KM Tran, ..., **N Rezaie**, et al. *Molecular Neurodegeneration* (2023).
- PyWGCNA: A Python package for weighted gene co-expression network analysis. **N Rezaie**, F Reese, A Mortazavi. *Bioinformatics* (2023).
- Somatic point mutations are enriched in non-coding RNAs with possible regulatory function in breast cancer. **N Rezaie**, ..., H Alinejad-Rokny. *Communications Biology* (2022).

Selected Talks and Posters

- Alzheimer's Association International Conference 2024. *Poster*: "5xFAD/Abi3^{S209F} mice display distinct effects on microglia dynamics."
- PARSE Biosciences 2023. *Invited Speaker*: "Identifying Distinct Cellular Programs Using Topyfic."
- Network Biology 2023 (CHSL). *Poster*: "Topyfic identified robust cellular programs using reproducible LDA."
- Alzheimer's & Dementia 2022 (AAIC). *Poster*: "Bulk and single-cell gene expression network analysis using PyWGCNA."
- Neuroscience 2020 (SFN). *Poster*: "Bulk and single-nucleus analysis of the 3xTgAD mouse model."

References

Available upon request.